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The preferences of laying hens for different concentrations of atmospheric ammonia

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Abstract

Ammonia gas is one of the most abundant aerial pollutants of modern poultry buildings. The current chronic exposure limit for ammonia of 25 ppm is set for human safety rather than animal welfare. This study assessed the behavioural preferences of laying hens (*Gallus gallus domesticus*) for different concentrations of ammonia found in commercial poultry houses. Six groups, each of six laying hens, were given the choice of three concentrations of ammonia (≈ 0 , 25 and 45 ppm) in a preference chamber over a period of 6 days and their location and behaviour recorded every 15 min. Hens foraged ($p = 0.018$), preened ($p = 0.009$) and rested ($p = 0.029$) significantly more in fresh air than in the ammonia-polluted environments. There was a significant difference between the responses in 0 and 25 ppm ($p < 0.05$) but not between 25 and 45 ppm ($p > 0.05$). This suggests that ammonia may be aversive to hens with a threshold for this aversion between 0 and 25 ppm. Future studies should explore graded concentrations of ammonia between 0 and 25 ppm in order to suggest a new chronic exposure limit on the basis of animal welfare. © 2000 Elsevier Science B.V. All rights reserved.

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1. Introduction

Ammonia is recognised as one of the most abundant aerial contaminants of poultry houses (Wathes et al., 1983). It is a colourless, highly irritant alkaline gas that is

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produced during the decomposition of organic matter (Anderson et al., 1964). Ammonia is water-soluble and can thus be absorbed in dust particles and litter as well as in mucus membranes, where it may cause intracellular damage (Vissek, 1968; Oyetunde et al., 1978; Al-mashhadani and Beck, 1983). Intensification of poultry production systems during the last decades has led to an increase in aerial pollutant emissions (Curtis and Drummond, 1982; Feddes and Licsko, 1993). The potential effects of poor air quality on poultry welfare involve complex interactions between physiology, behaviour and disease (Wathes, 1998). Despite overwhelming amounts of research on the effects of ammonia on the health and performance of poultry, no clear evidence is available currently to suggest whether poultry find ammonia aversive. Since animal welfare relates to suffering as a subjective experience, research into the behavioural effects of ammonia may assist the existing evidence and indicate whether the current exposure limits for poultry should be reevaluated.

1.1. Behavioural responses of livestock to ammonia

Exposure to ammonia may compromise poultry welfare by a variety of mechanisms, which can be related to FAWC's five freedoms (FAWC, 1992) (Table 1, Kristensen, 1998). Preference tests provide an important tool in animal welfare research (Nicol, 1986). In their principal form, animals are given a choice between two or more resources. The animal's choice may provide an insight into the way it perceives different resources in relation to other options (e.g. Hughes, 1977). Despite providing valuable information, preference tests have often been criticised, as summarised by Broom and Johnson (1993). Preference tests, in combination with various other behavioural techniques, have previously been applied to studies of animals' responses to ammonia.

Morrison et al. (1993) studied the aversion of pigs and poultry to various concentrations of ammonia but their findings were ambiguous, showing no consistent avoidance of ammonia by chickens. Subsequently, Smith et al. (1996) showed that pigs would overcome an initial spatial preference in order to avoid ammonia, suggesting an aversion. Atmospheric ammonia was also found to affect the behavioural repertoire of pigs by Jones et al. (1996). Over a period of 2 weeks, pigs preferred to rest, sit, feed and forage significantly more in an unpolluted environment when given the choice between approximately 0, 10, 20 and 40 ppm atmospheric ammonia. Pigs also spent most of their time (53.4%) in the unpolluted environments (Jones et al., 1996) although they would occupy ammonia-polluted environments to achieve both thermal comfort and companionship (Jones et al., 1999).

1.2. Aims and objectives

The current recommended chronic exposure limit for ammonia concentrations in poultry houses is 25 ppm (MAFF, 1987) although hourly concentrations exceeding 45 ppm are found in commercial poultry buildings (Groot Koerkamp et al., 1998). The recommended exposure limit in the UK is set by the Health and Safety Executive and is based on human safety rather than animal welfare (MAFF, 1987). Hence, research is needed to assess the effects of aerial pollutants, such as ammonia, on poultry, thus

Table 1
Evidence for the effects of ammonia on poultry for each of FAWC's five freedoms (Kristensen, 1998)

Freedom (FAWC, 1992)	Evidence for the effects of ammonia on poultry
(1) Freedom from hunger, thirst and malnutrition	Ammonia may reduce food intake in poultry and cause weight loss. Effects on thirst, feeding and drinking behaviour are not yet known.
(2) Freedom from discomfort	Ammonia causes irritation to mucus membranes which may cause discomfort.
(3) Freedom from pain, injury and disease	Ammonia causes air sac lesions, ketaro-conjunctivitis and increases susceptibility to many diseases. Rapid diagnosis of disease may be delayed due to ammonia aversion of the stockperson.
(4) Freedom to express normal behaviour	None of the reviewed research addresses the effects of ammonia on poultry behaviour.
(5) Freedom from fear and distress	No clear evidence is available on ammonia aversion, which could cause distress if the birds are unable to escape an aversive environment.

enabling an informed review of the current exposure limits from the point of animal welfare. The overall aim of this study was hence to assess the behavioural preferences of laying hens to different concentrations of atmospheric ammonia using a preference test. This may give an indication of whether poultry find ammonia aversive at concentrations to which they are exposed routinely in commercial poultry buildings.

2. Materials and methods

2.1. Animals and husbandry

Sixty ISA brown medium hybrid laying hens (*Gallus gallus domesticus*) were obtained from a commercial supplier at 16 weeks of age. They were housed in a mechanically ventilated room, measuring 4×4 m, and ring tagged for identification. The room contained litter (120 l. Snowflake Supreme Wood-shavings), a nesting area as well as twelve commercial nest boxes, a perch and various enrichment objects (e.g. a football, string, etc.), which were changed at regular intervals. The hens were offered a commercial diet, “Flavorlay pellets”, formulated to contain 17% protein, 3.9% fibre, 2.5% oil, 12.7% ash and 14% moisture. Food and water were available ad libitum at all times. The room was cleaned once weekly and fresh litter provided in order to avoid ammonia build up that could affect the experimental results.

2.2. Materials and experimental conditions

An environmental preference chamber was used in the experiment (Jones et al., 1996). It comprised eight identical compartments arranged in an octagonal annulus with access between adjacent compartments through doorways of adjustable height. Two groups of hens were tested simultaneously in three compartments on opposite sides of the chamber. The chamber was designed initially for studies on pigs and was modified to house hens in this experiment. A nesting area and a suspended feeder were provided in each compartment and the windows between compartments were blanked. A video camera was fitted above the transparent roof in each compartment. The doorways between adjacent compartments were fitted with plastic curtains through which the hens could move freely. The design of the chamber allowed each compartment to be polluted with ammonia gas independently of all other compartments.

2.2.1. Ammonia supply to the chamber

The system for the supply of ammonia gas was modified from that described by Jones et al. (1996) in order to improve control of the desired concentrations of ammonia in the chamber compartments. Ammonia gas was supplied from a cylinder of compressed anhydrous ammonia and its flow adjusted by a regulator to a pressure of 1 bar. Primary dilution was provided by compressed air, which was supplied at a pressure of 4.2 bar and a flow rate of 70 l/min. Further dilution was achieved by the division of this primary ammonia supply into 32 secondary supply lines, each governed by an orifice to an airflow of 2.37 ± 0.03 l/min. Flow rates were measured using a SAGA digital flow

meter. Each secondary supply line would thus, when connected to the main ventilation manifold to a particular compartment, raise the ammonia concentration by approximately 11–12 ppm. Up to four secondary supply lines could be connected to each of the eight compartments in the chamber, allowing a maximum ammonia concentration of approximately 48 ppm. The desired concentration in any particular compartment was adjusted by connecting the appropriate number of secondary supply lines to the ventilation inlet. Those secondary supply lines, which were not in use, were connected to the ventilation outlet hence bypassing the chamber completely. The safety cut-out mechanisms and ventilation system of the chamber were controlled in a similar manner to that described by Jones et al. (1996) with an overall air change rate of approximately 57 per hour.

The ammonia concentration in each compartment was monitored throughout the duration of the experiment. Gas samples were collected continuously via a multiplexer bypass pump arrangement, which switched between the sampling lines from each compartment's outlet every 7 min. The sample was then pumped via an ammonia converter (stainless steel catalyst at 750°C) to a chemiluminescence NO_x analyser (model 42I, Thermo Environmental Instruments). A data logger controlled the sampling sequence and recorded the ammonia concentration from the analyser throughout the experiment.

Three ammonia concentrations of nominally 0, 25 and 45 ppm were used in the study. The control concentration was as close to 0 ppm as could be achieved with an average concentration during the experiment of 1.5 ppm ammonia. During the experiment, the ammonia concentrations in the chamber varied by approximately 10% with no overlap between the set concentrations.

The preference chamber was modified further in order to ensure even light intensities between the compartments since preferences for particular light intensities in poultry have been shown previously (Davis et al., 1999). Six luminaires, each with a 60 W light bulb, were placed on three layers of tracing paper on the top of each compartment. Five of the luminaires were set on a 12:12 h photoperiod providing approximately 99 ± 1.88 lux from 0600 to 1800 h. A single luminaire on a central dimmer switch ran continuously, providing dim light of approximately 2.75 ± 0.25 lux during the night for identification of the hens from the video images. The light intensities were measured using a TES-1334 light meter, sited 150 mm above the floor and averaged over four locations within each compartment. Temperature and relative humidity were monitored in each of the compartments during the experiment and varied between compartments by approximately 2°C and 5% respectively.

The hens were fitted with harnesses made from bright green lycra, which were marked in unique patterns with black stockmarker for individual identification. The harnesses allowed the hens to perform all normal behaviours including dustbathing and wing stretching.

2.3. Experimental procedures

The birds were introduced to the preference chamber in six groups, each of six hens, which were randomly selected from the main flock of sixty (every fifth bird encountered

was selected). Two groups were tested simultaneously in opposite sides of the preference chamber. No hen was used more than once. Each hen was weighed, fitted with an identification harness and randomly allocated to a starting compartment to which it was returned at the beginning of each treatment period. Each group of hens had free access to three compartments during an 8-day period. This comprised 2 days for acclimatisation to the chamber followed by three treatment periods, each lasting 2 days. The location and behaviour of the hens were recorded continuously by time-lapse video. The three concentrations of ammonia (nominally 0, 25 or 45 ppm) were applied to the three compartments in a latin square design and maintained for the 48-h duration of each treatment period. Hence, by the end of the experiment, all compartments had contained all concentrations of ammonia. The 2-day period was chosen to mimic the chronic exposure that hens may experience in commercial production and to overcome any exploratory or patrolling behaviour that might have compromised the assumptions of a preference test. Eggs were collected every day from outside the chamber to minimise the disturbances to the hens. At the end of each 2-day treatment period, the ammonia supply was switched off and hens removed from the chamber. The drinkers were cleaned and food replenished. The ammonia supply lines were reconnected to the appropriate compartment inlets and the hens returned to their allocated starting compartments.

2.4. *Recordings and statistical analyses*

The behaviour and location were recorded for each hen on a 15-min instantaneous sampling interval for the three treatment periods for each group of six hens. Predefined, mutually exclusive behavioural categories were identified from an ethogram (Table 2) to minimise recording errors. The sampling technique and interval were chosen to capture the behaviour of the hens at regular intervals throughout the exposure period, although this sampling method may underestimate infrequently occurring behaviours.

The data were processed using Excel 7.0 and summarised for each group of hens in each of the ammonia concentrations for each of the 2-day treatment periods. Further

Table 2
Ethogram of recorded behaviours

Behavioural category	Definition
Drink	The beak is situated in or above the water trough.
Eat	The head is situated less than approximately 5 cm from the suspended feeder with the beak pointing towards the feeder.
Forage	The hen is pecking or scratching at the substrate.
Nest	The hen is situated within the nest box.
Other	None of the otherwise described behaviours including out-of-sight.
Peck	Sharp forward movements with the head towards an object or conspecific.
Preen	The beak is moving whilst touching another part of the body of the hen.
Rest	Stationary — either standing or sitting.
Dust bathe	A complex behavioural repertoire identified by the hen lying on its side displaying at least one leg to the side.
Walk	Moving one foot in front of the other resulting in forward movement of the hen.

statistical analyses were performed using Genstat 5, release 3.2 for windows. The data were tested for normality, constancy of variances and additivity of effects between treatments. Variables, which violated the above assumptions, were either log-or square root transformed, according to their distribution. An ANOVA was used to investigate each of the behaviours in relation to the ammonia concentrations and also to assess any interactions with other factors, such as light intensity. Post hoc *t*-tests were carried out on the significantly affected behaviours in order to discriminate between the effects of the individual concentrations. A residual maximum likelihood (REML) analysis was adopted as an extension of the ANOVA method to assess the effects of compartment preferences and ammonia treatment on egg-production. The duration of visits to the different ammonia concentrations was assessed by summing the number of consecutive counts in each of the concentrations for individual birds over the exposure period. The frequency of visits with a specific duration was calculated and subjected to a χ^2 analysis in order to assess whether any differences in the distribution of the duration of visits were due to the different ammonia concentrations.

3. Results

3.1. Behavioural time budget of the hens

The hens spent most of their time foraging (30.3%), resting (24.6%) and preening (14.3%). These behaviours made up 69.2% of the total time budget as shown in Fig. 1. The relative occurrences of individual behaviours varied significantly irrespective of ammonia concentration (ANOVA, $F_{(9, 459)} = 70.69$, $p < 0.001$, S.E.D. = 0.0637). The effects of ammonia on the overall behavioural repertoire of the hens were assessed by adopting “behavioural category” as a second treatment factor in the ANOVA model. No significant interactions were found between the proportions of behaviour

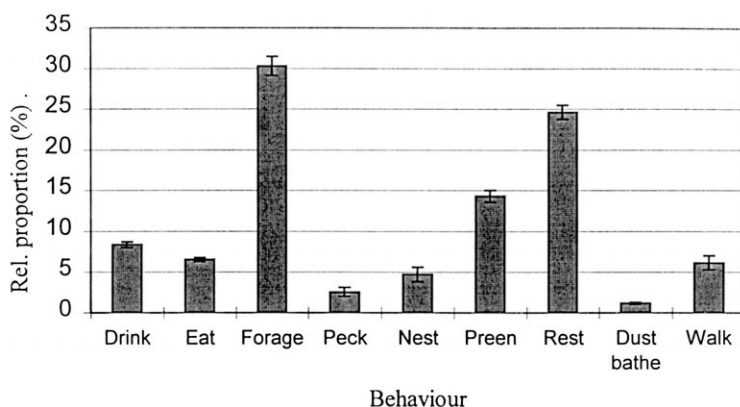


Fig. 1. The relative proportions of behaviours averaged over the groups during the experiment.

Table 3
The mean occurrences of behaviour per 2-day treatment period in the different ammonia concentrations

Behavioural category	Ammonia concentration (ppm)						$F_{(2,22)}$	S.E.D.
	0		25		45			
	Mean	S.E.	Mean	S.E.	Mean	S.E.		
Drink	37.8	7.78	28.6	7.93	28.0	8.34	2.00	5.53
Eat	28.9	5.92	20.6	6.20	24.5	7.19	1.46	4.85
Forage	146.4	31.20	103.5	28.50	91.4	27.00	4.85 *	18.54
Nest ^a	22.6 (1.2)	– 0.08	17.7 (1.0)	– 0.12	15.8 (1.0)	– 0.13	– 1.41	– 0.14
Peck	12.2	3.64	7.4	2.52	9.3	2.81	1.85	2.50
Preen	70.2	17.30	49.0	14.20	42.4	13.60	5.81 * *	8.53
Rest ^a	121.1 (1.6)	– 0.20	70.3 (1.1)	– 0.23	89.1 (1.3)	– 0.23	– 4.19 *	– 0.19
Dustbathe	4.4	1.50	5.2	1.66	3.6	1.00	0.80	1.23
Walk	26.1	4.61	22.0	5.23	21.3	3.95	0.82	4.06
Total ^a	474.2 (2.4)	– 0.15	326.3 (2.1)	– 0.17	330.3 (2.1)	– 0.15	– 5.14 *	– 0.10

– Not applicable.

* $p < 0.05$.

^aLog-transformed variable. The back-transformed means are given with the transformed mean in parentheses and their equivalent S.E., F -value and S.E.D.

** $p < 0.01$.

(log-transformed) and the ammonia concentrations (ANOVA, $F_{(18, 459)} = 1.00$, $p = 0.457$, S.E.D. = 0.133).

3.2. The effects of ammonia on individual behaviours

The occurrences of each behaviour category in the three ammonia concentrations were summed over each 2-day treatment period for every group. A separate ANOVA was performed for each behavioural category. Ammonia concentration was found to have a significant effect on the amount of foraging, resting and preening behaviour as well as on the total occupancy of any environment (Table 3). Although the hens were observed more frequently in 0 ppm than in 25 and 45 ppm, the hens showed individual variation in their occupancy of the different ammonia concentrations (Fig. 2). This

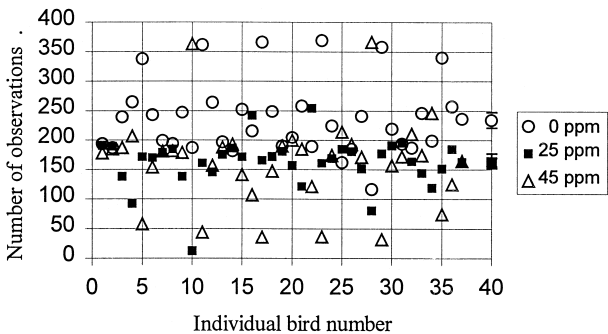


Fig. 2. The number of observations in different ammonia concentrations for individual hens over 2 days. The mean occupancies are included with standard errors as bird number 40.

individual variation in occupancy was higher in 45 ppm than in the lower concentrations.

A post hoc *t*-test revealed that all of the significantly affected behaviours occurred more often in fresh air than in 25 ppm ammonia ($t = 2.31$ – 3.08 , $df = 22$, $p < 0.05$). There were no significant differences between the occurrences of behaviours in the two ammonia-polluted environments ($t = 0.65$ – 0.80 , $df = 22$, $p > 0.05$). No significant interactions were found between the light level and ammonia concentration for any of the recorded behaviours.

3.3. The preferences for a nesting site

Each hen laid an average of 0.98 eggs/day. The number of eggs laid did not vary significantly between the different ammonia concentrations (log-transformed, ANOVA, $F_{(2, 22)} = 1.75$, $p = 0.197$, S.E.D. = 0.085).

3.4. Visit duration

The most frequent visit duration was 0–15 min (data not shown). The distribution of the duration of visits lasting between 0 and 150 min depended significantly on the concentration of ammonia ($\chi^2 = 30.72$, $df = 18$, $p = 0.031$). Inspection of the data showed that the frequency of visits longer than 75 min was higher in fresh air than in 25 or 45 ppm ammonia. Table 4 shows the overall frequency of visits of long durations in the different ammonia concentrations, indicating that short visits were more frequent than long visits and that more visits were made to the lower concentrations of ammonia. There was a significant difference in the distribution of visit durations between ammonia concentrations ($\chi^2 = 34.18$, $df = 20$, $p = 0.025$).

Table 4
Overall frequency of visit durations in the different ammonia concentrations for 36 hens

Visit duration (min)	Ammonia concentration (ppm)			
	0	25	45	Total
0–150	439 ^a	421	416	1276
151–300	62	44	28	134
301–450	16	12	14	42
451–600	18	9	3	30
601–750	17	11	14	42
751–900	9	9	6	24
901–1050	17	8	10	35
1051–1200	12	5	7	24
1201–1350	8	4	6	18
1351–1500	11	4	4	19
1501–2880	2	7	7	16
Total	611	534	515	1660

^aTotal number of visits of the specified duration made by hens to a particular concentration of ammonia over 2 days.

4. Discussion

4.1. *The effect of ammonia on the behaviour of laying hens*

The hens spent significantly more time foraging, resting and preening in fresh air than in the ammonia-polluted environments. These behaviours made up 69.2% of the total time budget, which in turn was significantly affected by ammonia concentration.

Previous studies indicate that feeding is affected by high concentrations of ammonia (Quarles and Kling, 1974; Johnson et al., 1991; Emeash et al., 1998). Despite no differences in the amount of eating behaviour between the aerial environments, we found significantly less foraging behaviour in the ammoniated environments than in the fresh air. Hence, the results may reflect the distinction between “eating” and “foraging” in the ethogram.

The behavioural effects of ammonia may have several potential causes. As a water soluble gas, ammonia can be absorbed into the mucus membranes and cause damage to eyes (kerato-conjunctivitis) as well as to the respiratory system, both of which may be painful to the birds (Cohen and Gold, 1975; Oyetunde et al., 1978; Al-mashhadani and Beck, 1983). These physiological effects could lead to a reduced or altered sensory input from the environment, which may in turn affect many behaviour patterns. More specific studies determining the causes of these effects of ammonia are needed, not only because the animals may be suffering pain, discomfort or hunger but also since production may be affected in the longer term for both layers and broilers.

The amount of preening was significantly affected by ammonia concentration. The reasons for this are less obvious. Since preening involves physical contact of the head with the surface of the feathers, ammonia-containing feathers may have acquired an aversive taste or smell. The inhibition of normal behaviour patterns, such as grooming or preening, has been suggested as an indicator of compromised welfare (Broom and Johnson, 1993). Hence, it is plausible that the reduction in preening behaviour found in ammonia-polluted environments is indicative of an aversion to ammonia.

Significant differences were discovered between the behaviour of hens in fresh air and 25 ppm but not between 25 and 45 ppm. This suggests that there may be a threshold for ammonia aversion between 0 and 25 ppm. MAFF's recommended chronic exposure limit for human safety in poultry houses is currently 25 ppm (MAFF, 1987). Further studies of the behavioural responses of poultry to ammonia should examine the concentrations of ammonia between 0 and 25 ppm in order to suggest a more appropriate chronic exposure limit for poultry houses in terms of animal welfare.

4.2. *Visit duration*

The assessment of the length of visits to the different ammonia concentrations allowed more specific evidence for an ammonia aversion to be made from the study. The frequency of visits lasting less than 75 min was similar for all ammonia concentrations. This corresponds to previous findings of exploratory and patrolling behaviour in hens and confirms that the hens had experienced all ammonia concentrations in the preference test. A common limitation of preference tests is that the first environment

chosen by the animal is noted as the preferred one (Broom and Johnson, 1993). An assessment of visit duration can provide information on an animal's motivation to exit rather than to enter a specific environment, hence appreciating the natural exploratory behaviour of most animal species. The distribution of visits lasting longer than 75 min differed significantly between the environments. This suggests that the hens were able to discriminate between the environments after 75 min ammonia exposure. A delayed aversion to ammonia has been shown in pigs (Jones et al., 1996), though this is the first evidence of a delayed aversion to ammonia in poultry. The delay could be related to the buffer capacity of ammonia in the epithelial cells of mucus membranes since toxicity depends on penetration of cell membranes (Visec, 1968). However, the findings need confirmation from future studies using alternative sampling techniques and more graded concentrations of ammonia between 0 and 25 ppm.

4.3. The effects of ammonia on the production of laying hens

The number of eggs laid did not vary significantly between the ammonia concentrations. This suggests either a strong motivation to nest irrespective of environmental conditions or the presence of a spatial preference for a nest site that was stronger than the aversion to ammonia. Millam (1987) found that turkey hens preferred to nest in boxes at one end of a row compared to those in the middle. Appleby et al. (1986) found that laying hens preferred nesting in the diagonal corners of a deep litter house.

4.4. Critique of experimental methods

The design of the preference chamber allowed several of the traditional limitations of preference tests to be overcome. Longer-term preferences could be assessed since each compartment provided identical facilities so that the hens could be kept in the chamber for the duration of the experiment. One of the criticisms of preference testing is that the choices are non-exclusive and that the minority choices are ignored when interpreting the results (Dawkins, 1980; Nicol, 1986). This study was designed to accommodate the non-exclusive preferences for ammonia concentrations by assessing not only the location of the hen but also the behaviour within that location in relation to other concentrations. Further, the length of the experimental period and the assessments of the visit durations may have overcome some of the limitations of conventional preference tests. However, preference testing cannot provide evidence of the preferred environment in relation to all other environmental factors and cannot solely provide evidence for the strength of an aversion. Further studies, adopting different behavioural techniques, should be carried out in order to assess the findings of this study in relation to other social and environmental factors as well as attempting to explain the underlying mechanisms for the behavioural effects found in this study.

On the basis of this study, it is concluded that laying hens, when given a free choice, prefer to spend more time in fresh air than in ammonia-polluted environments. The evidence for behavioural changes from this study suggests a threshold for aversion to ammonia at or below 25 ppm, which should be investigated further in order to improve the welfare of laying hens in the future.

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